Light

GCSE Computer Science — Networks Basics (Set 1)

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GCSE • Edexcel • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (Edexcel)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

A school connects all the computers in its building using cables and switches.
What type of network is this?

- **A.** A Local Area Network (LAN)
- **B.** A Wide Area Network (WAN)
- **C.** The Internet
- **D.** The World Wide Web

Q2 (1 mark)

A company has offices in London, New York, and Tokyo. All offices are connected so employees can share files.
What type of network connects these offices?

- ○ **A.** A Local Area Network (LAN)
- ○ **B.** A Wide Area Network (WAN)
- ○ **C.** A Personal Area Network (PAN)
- ○ **D.** A Bluetooth network

Q3 (1 mark)

In a network, one powerful computer stores all the files and manages user logins. Other computers request data from it.
What type of network model is this?

- ○ **A.** Client-server
- ○ **B.** Peer-to-peer
- ○ **C.** Standalone
- ○ **D.** Mesh

Q4 (1 mark)

Four friends connect their laptops to share music files. Each laptop can access files stored on any other laptop. There is no central computer controlling the network.
What type of network model is this?

- ○ **A.** Client-server
- ○ **B.** Peer-to-peer
- ○ **C.** Cloud computing

- ○ **D.** Mainframe

Q5 (1 mark)

What is the Internet?

- ○ **A.** A collection of web pages linked by hyperlinks
- ○ **B.** A global network of interconnected computer networks
- ○ **C.** A web browser used to view websites
- ○ **D.** A search engine for finding information

Q6 (1 mark)

What is the World Wide Web (WWW)?

- ○ **A.** The physical cables that connect computers globally
- ○ **B.** The same thing as the Internet
- ○ **C.** A type of network cable
- ○ **D.** A collection of web pages accessed via the Internet using browsers

Q7 (1 mark)

A business decides to connect all its computers in a network.
Which is a benefit of networking computers?

- ○ **A.** Each computer needs its own printer
- ○ **B.** Files can be easily shared between computers
- ○ **C.** Software must be installed separately on every computer
- ○ **D.** Users cannot communicate with each other

Q8 (1 mark)

Which is a potential drawback of connecting computers to a network?

- ○ **A.** Users can share files easily
- ○ **B.** Printers can be shared between users
- ○ **C.** Malware can spread more easily between computers
- ○ **D.** Communication between users is improved

Q9 (1 mark)

Who typically owns and manages the hardware in a Local Area Network (LAN)?

- ○ **A.** The Internet Service Provider (ISP)
- ○ **B.** The government
- ○ **C.** A global telecommunications company
- ○ **D.** The organisation that uses the network

Q10 (1 mark)

What does 'bandwidth' measure in a computer network?

- ○ **A.** The physical width of network cables
- ○ **B.** The maximum amount of data that can be transmitted per second
- ○ **C.** The number of computers connected to the network
- ○ **D.** The distance between two computers

Q11 (1 mark)

In a client-server network, what is the role of the server?

- ○ **A.** To provide services and resources to client computers
- ○ **B.** To request services from other computers
- ○ **C.** To connect to the Internet only
- ○ **D.** To act as a backup power supply

Q12 (1 mark)

In a client-server network, what is the role of a client?

- ○ **A.** To store all the network's files centrally
- ○ **B.** To manage user accounts and permissions
- ○ **C.** To request services and data from the server

- ○ **D.** To connect different networks together

Q13 (2 marks)

A student says: 'A WAN always uses infrastructure owned by third parties like telecommunications companies.'
Is this statement correct for most WANs?

- ○ **A.** No, WANs always use the organisation's own cables
- ○ **B.** No, WANs only exist within single buildings
- ○ **C.** Yes, WANs typically lease connections from telecommunications providers
- ○ **D.** Yes, but only for international connections

Q14 (2 marks)

Which is an advantage of a client-server network over peer-to-peer?

- ○ **A.** No server hardware is required
- ○ **B.** Easier to set up with no technical knowledge
- ○ **C.** Centralised backup and security management
- ○ **D.** All computers must be switched on to access files

Q15 (2 marks)

Which is an advantage of peer-to-peer over client-server?

- ○ **A.** Better security and centralised user management
- ○ **B.** Lower cost as no dedicated server is needed
- ○ **C.** Easier to manage with hundreds of users
- ○ **D.** Files are backed up automatically to a central location

Q16 (2 marks)

A company's network becomes slow when many employees are using it at the same time.
What is this problem called?

- ○ **A.** Bandwidth
- ○ **B.** Latency
- ○ **C.** Encryption
- ○ **D.** Network congestion

Q17 (2 marks)

A gamer complains about 'lag' when playing online games.
Which network factor most directly causes this problem?

- ○ **A.** High bandwidth
- ○ **B.** High latency
- ○ **C.** Strong encryption
- ○ **D.** Large file sizes

Q18 (2 marks)

A student says: 'Email uses the Internet but is not part of the World Wide Web.'
Is this statement correct?

- ○ **A.** No, email is part of the World Wide Web
- ○ **B.** No, email does not use the Internet
- ○ **C.** Yes, email uses Internet protocols but is a separate service from the WWW
- ○ **D.** Yes, but only web-based email like Gmail

Q19 (2 marks)

An office has 20 computers and one printer connected to a network.
Which statement correctly describes a benefit of this setup?

- ○ **A.** All 20 computers can share the single printer
- ○ **B.** Each computer needs its own dedicated printer
- ○ **C.** The printer can only be used by one computer
- ○ **D.** Printing is not possible over a network

Q20 (2 marks)

A company's network is connected to the Internet.
Which security risk does this introduce?

- ○ **A.** Employees can share files internally
- ○ **B.** The network will have higher bandwidth

- ○ **C.** Computers will run faster
- ○ **D.** Hackers could potentially access the network remotely

Q21 (2 marks)

Which statement about Wide Area Networks (WANs) is correct?

- ○ **A.** WANs are faster than LANs because they cover more distance
- ○ **B.** WANs are typically slower than LANs due to the distances involved
- ○ **C.** WANs can only connect two locations
- ○ **D.** WANs do not require any specialised equipment

Q22 (2 marks)

Which transmission medium typically provides the highest bandwidth for network connections?

- ○ **A.** Fibre optic cables
- ○ **B.** Copper cables
- ○ **C.** Wi-Fi signals
- ○ **D.** Bluetooth

Q23 (2 marks)

In a client-server network, what happens if the server fails?

- ○ **A.** Clients can continue working with full functionality
- ○ **B.** The network automatically switches to peer-to-peer
- ○ **C.** Nothing, because data is stored on all computers
- ○ **D.** Clients lose access to centralised services and data

Q24 (3 marks)

A new business is choosing between peer-to-peer and client-server for their network of 5 employees.
They have a limited budget but need to share files.
Which network model is most appropriate and why?

- ○ **A.** Client-server, because it provides better security for all network sizes
- ○ **B.** Peer-to-peer, because it's cheaper and sufficient for a small number of users
- ○ **C.** Client-server, because peer-to-peer cannot share files
- ○ **D.** Neither, because 5 users is too few for any network

Q25 (3 marks)

A company experiences slow network performance. Investigation reveals:
• 100 Mbps bandwidth
• 500 users active simultaneously
• Large video files being transferred
• Old network switches
Which combination of factors is most likely causing the slowdown?

- ○ **A.** Only the bandwidth limitation

- ○ **B.** Only the number of users
- ○ **C.** The combination of many users, large files, and possibly outdated hardware
- ○ **D.** Old switches have no effect on performance

Q26 (3 marks)

A student lists services that 'use the Internet':
1. World Wide Web (websites)
2. Email
3. Online gaming
4. Video streaming
5. File Transfer Protocol (FTP)

How many of these services use the Internet but are NOT part of the World Wide Web?

- ○ **A.** 1 service (only email)
- ○ **B.** 2 services (email and FTP)
- ○ **C.** 3 services (email, gaming, and FTP)
- ○ **D.** 4 services (all except WWW)

Q27 (3 marks)

A hospital is implementing a new computer network.
Consider these statements:
1. Patient records can be accessed from any department
2. A virus could potentially spread to all connected systems
3. Centralised backup protects against data loss
4. Network failure could affect critical systems

Which statements describe benefits and which describe drawbacks?

- ○ **A.** All statements are benefits

- ○ **B.** Statements 1 and 3 are benefits; 2 and 4 are drawbacks
- ○ **C.** Statements 1 and 2 are benefits; 3 and 4 are drawbacks
- ○ **D.** All statements are drawbacks

Q28 (3 marks)

A university has:
• A main campus in Manchester
• A research facility in Edinburgh
• Student accommodation buildings adjacent to the main campus
Which statement correctly describes the network types needed?

- ○ **A.** The main campus and accommodation need LANs; a WAN connects to Edinburgh
- ○ **B.** One large LAN can connect all three locations
- ○ **C.** All three locations require separate WANs
- ○ **D.** WANs are used within buildings; LANs connect different cities

Q29 (3 marks)

A growing company currently uses peer-to-peer networking for 8 employees.
They plan to expand to 50 employees next year.
Which statement best evaluates their network needs?

- ○ **A.** Peer-to-peer will continue to work well because it scales easily
- ○ **B.** They should switch to client-server for better management, security, and scalability
- ○ **C.** They should keep peer-to-peer because client-server is only for 1000+ users

- ○ **D.** Network type doesn't matter with more employees

Q30 (3 marks)

A school's network is experiencing slow performance.
Which action would most likely improve network speed?

- ○ **A.** Adding more users to the network
- ○ **B.** Upgrading from 100 Mbps to 1 Gbps switches and cabling
- ○ **C.** Removing the firewall to reduce security overhead
- ○ **D.** Converting from client-server to peer-to-peer

Total: **55** marks • Time: **30 min**

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